



Pectopteris fern

CARBONIFEROUS:
359–299 MYA

Before the evolution of flowering plants, forests abounded with spore-bearing species, including ferns like *Pectopteris*.



Ichthyosaurus

JURASSIC: 200–145 MYA

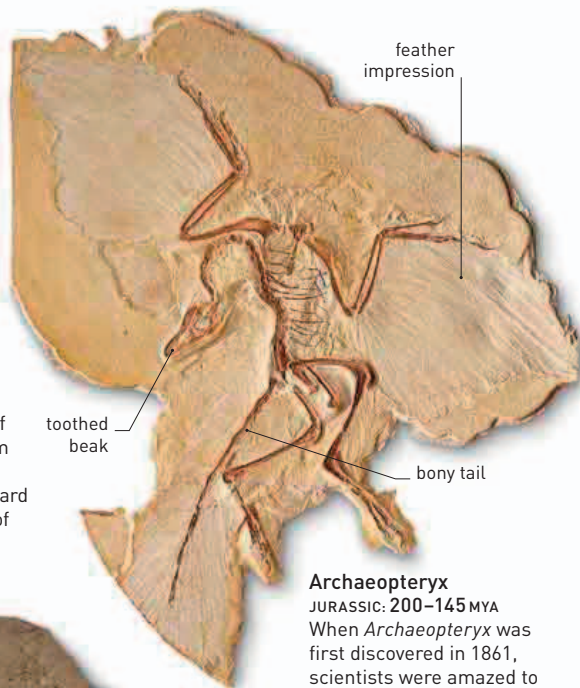
In 1811, fossil hunter Mary Anning collected the first complete skeleton of an ichthyosaurus, a predatory marine reptile. Ten years later, she found the first plesiosaur.



Diplomystus dentatus

PALEOGENE: 65–23 MYA

Fossils of an early form of modern-day herring, from Wyoming, were first described in 1877 by Edward Drinker Cope, a pioneer of American paleontology.



Archaeopteryx

JURASSIC: 200–145 MYA

When *Archaeopteryx* was first discovered in 1861, scientists were amazed to see impressions of feathers next to sharp teeth and a bony tail. It was a missing link between reptiles and birds.



fossilized bark

Fossilized wood

TRIASSIC: 251–200 MYA

Dead matter, such as wood, fossilizes when it is changed into mineral. The 20th century's radiometric dating allowed scientists to determine the age of fossil-bearing rock.

Dinosaur footprints

JURASSIC: 200–145 MYA

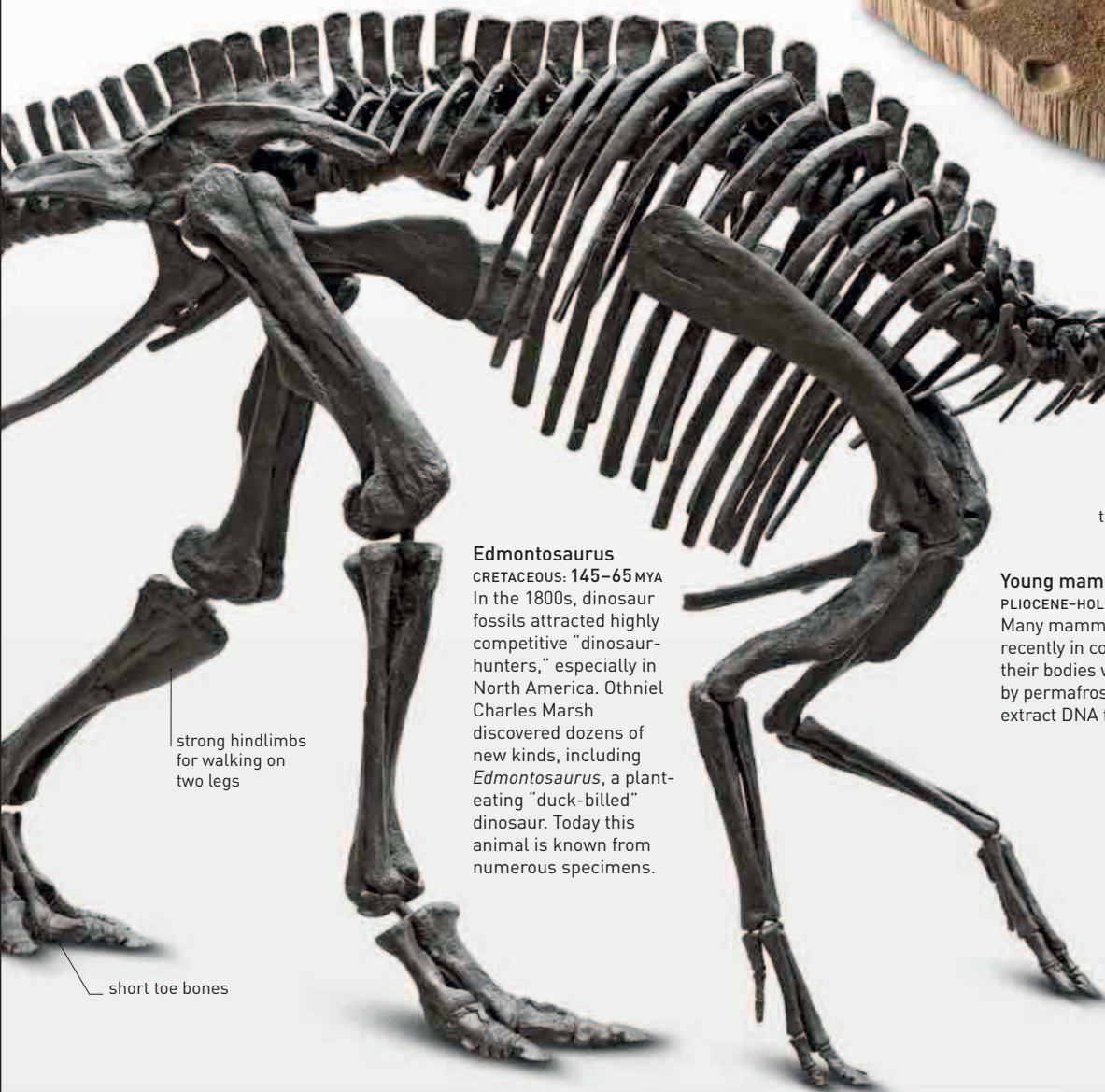
Trace fossils, such as footprints, can reveal aspects of animal behavior. Here, prints of the predatory *Allosaurus* record its pursuit of prey.



Allosaurus prints



skull with ducklike "bill"



Edmontosaurus

CRETACEOUS: 145–65 MYA

In the 1800s, dinosaur fossils attracted highly competitive "dinosaur-hunters," especially in North America. Othniel Charles Marsh discovered dozens of new kinds, including *Edmontosaurus*, a plant-eating "duck-billed" dinosaur. Today this animal is known from numerous specimens.

strong hindlimbs for walking on two legs

short toe bones



thick skin

Young mammoth in permafrost

PLIOCENE-HOLOCENE: 5.3 MYA–4,500 YA

Many mammoths lived comparatively recently in cold Siberian landscapes, so their bodies were sometimes "mummified" by permafrost. Modern biologists can extract DNA from such specimens.



heavy brow ridge

Neanderthal skull

PLEISTOCENE: 2.5 MYA–12,000 YA

Although first unearthed in 1829, this kind of skull was not seen as evidence for another human species until others were found in the Neander Valley, Germany, in 1856.